

Tricky stuff

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0.1 Measurable set that is not borel

Let $\mathcal{C} \subset [0, 1]$ be Cantor set, $c: [0, 1] \rightarrow [0, 1]$ be cantor function defined by

$$\begin{aligned} c &= \frac{1}{2} \text{ on } \left[\frac{1}{3}, \frac{2}{3} \right] \\ c &= \frac{1}{4} \text{ on } \left[\frac{1}{9}, \frac{2}{9} \right], \quad c = \frac{3}{4} \text{ on } \left[\frac{7}{9}, \frac{8}{9} \right] \\ &\vdots \\ c \left(\sum_{n \in \mathbb{N}} \frac{2a_n}{3^n} \right) &= \sum_{n \in \mathbb{N}} \frac{a_n}{2^n} \quad \forall \{a_n\}_{n \in \mathbb{N}}, \quad a_n \in \{0, 1\} \\ c(x) &= \sup_{y \leq x} c(y) \text{ for } x \in [0, 1] \setminus \mathcal{C} \end{aligned}$$

Then c is continuous, surjective and nondecreasing. Therefore

$$\psi(x) = c(x) + x$$

is continuous, increasing and bijective $[0, 1] \rightarrow [0, 2]$. Since

$$\begin{aligned} [0, 1] \setminus \mathcal{C} &= \bigcup_{n \in \mathbb{N}} (a_n, b_n) \text{ where } (a_n, b_n) \text{ are intervals} \\ \lambda(\psi(a_n, b_n)) &= \lambda(c(a_n) + a_n, c(b_n) + b_n) = \lambda(c|_{A_n} + a_n, c|_{A_n} + b_n) = \lambda(a_n, b_n) \\ \lambda(\psi([0, 1] \setminus \mathcal{C})) &= \lambda([0, 1] \setminus \mathcal{C}) = 1 \\ \lambda(\psi(\mathcal{C})) &= \lambda(\emptyset) \end{aligned}$$

On the other hand, the completion of Borel sigma algebra is easily seen to be Lebesgue sigma algebra:

Denote $\mathcal{B}$ Borel algebra, $\mathcal{L}$ Lebesgue algebra, $\overline{\mathcal{B}}^\lambda$ the completion of $\mathcal{B}$ w.r.t. Lebesgue measure λ , G_δ to be set of countable intersections of open sets ("G $_\delta$ " sets). Clearly $\overline{\mathcal{B}}^\lambda \subset \mathcal{L}$. To prove the other inclusion, choose arbitrary $A \in \mathcal{L}$. Then $\exists B \in G_\delta: A \subset B, \lambda(B \setminus A) = 0$, so

$$A = \underbrace{B}_{\in \mathcal{B}} \setminus \underbrace{(B \setminus A)}_{\in \overline{\mathcal{B}}^\lambda, \text{ since } \lambda(B \setminus A) = 0} \in \overline{\mathcal{B}}^\lambda$$

0.2 Hölder continuous function not absolutely continuous

Let c be the cantor function and c_k be defined by

$$c_k \left(\sum_{n=1}^k \frac{2a_n}{3^n} \right) = \sum_{n=1}^k \frac{a_n}{2^n} \quad \forall \{a_n\}_{n=0}^k, a_n \in \{0, 1\}$$

c_k is piecewise linear on $[0, 1]$

$$c_k(0) = 0, c_k(1) = 1$$

Then

- $\|c - c_k\|_\infty = 2^{-k}$
- c_k is Lipschitz with constant $\left(\frac{3}{2}\right)^k$ (think about the slope of steepest line)

So for $x, y \in [0, 1]$

$$\begin{aligned} |c(x) - c(y)| &\leq \|c - c_k\|_\infty + |c_k(x) - c_k(y)| + \|c - c_k\|_\infty \\ &\leq 2 \cdot 2^{-k} + \left(\frac{3}{2}\right)^k |x - y| \end{aligned}$$

Choose $k \in \mathbb{N}$ such that $3^{-k-1} \leq |x - y| < 3^{-k}$, then

$$|c(x) - c(y)| \leq 3 \cdot 2^{-k} = 6 \cdot 2^{-k-1} = 6 \cdot 3^{-(k-1) \cdot \log_3 2} < 6 \cdot |x - y|^{\log_3 2}$$

So the cantor function is $\log_3 2$ -Hölder continuous.

Let $\epsilon > 0$. Since c is locally constant on $[0, 1] \setminus \mathcal{C}$, which is a countable union of open intervals with measure one, by continuity of measure we can choose $x_1, \dots, x_n, y_1, \dots, y_n \in [0, 1], x_k < y_k$ (in fact they will be endpoints of form $\frac{l}{3^n}, l, n \in \mathbb{N}$) such that

$$\sum_{k=1}^n |y_k - x_k| > 1 - \epsilon \quad \text{and } c \text{ is constant on intervals } (x_k, y_k)$$

Since c is constant on big set, and $c(0) = 0, c(1) = 1$, it must grow fast on a small set:

$$|x_1 - 0| + \sum_{k=1}^{n-1} |x_{k+1} - y_k| + |1 - y_n| = 1 - \sum_{k=1}^n |y_k - x_k| < \epsilon$$

$$\text{BUT } c(x_1) - c(0) + \sum_{k=1}^{n-1} (c(x_{k+1}) - c(y_k)) + c(1) - c(y_n) = 1 - \sum_{k=1}^n (c(y_k) - c(x_k)) = 1$$

so c is not $AC[0, 1]$. A shorter proof might use characterization $AC[0, 1] = W^{1,1}(0, 1)$, since c has classical derivative zero almost everywhere, but certainly

$$c(0) + \int_0^1 c'(x) dx = 0 + \int_0^1 0 dx = 0 \neq c(1)$$

In fact, $c \in BV[0, 1] \setminus AC[0, 1]$ and it's Lebesgue-Stieltjes measure is singular nonatomic with respect to the Lebesgue measure on $[0, 1]$.

0.3 Hopf fibration, dirac belt

$$S^1 \rightarrow S^3 \cong B_{1,\mathbb{C}^2}(0) \rightarrow S^2 \cong \mathbb{P}\mathbb{C}^1$$

Hilbert space for two-state system is

$$\mathcal{H} = \mathbb{C}^2 = \mathbb{C} |\uparrow\rangle \oplus \mathbb{C} |\downarrow\rangle$$

When we mod out the norm, we are left with $B_{1,\mathbb{C}^2}(0) \cong S^3$ and when we mod out the phase, we are left with $\frac{S^3}{S^1} \cong S^2$. Quantum evolution happens on S^3 and state space is $S^2 = \mathbb{P}(\mathcal{H})$.

Below we use pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \vec{\sigma} = \begin{pmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \end{pmatrix}$$

The Hopf fibration is the map which sends spinors (elements of S^3) to the expectation values of spin in the x, y, z directions:

$$\begin{aligned} \langle \bullet | \vec{\sigma} | \bullet \rangle : S^3 &\rightarrow S^2 \\ \alpha |\uparrow\rangle + \beta |\downarrow\rangle = |\psi\rangle &\mapsto \langle \psi | \vec{\sigma} | \psi \rangle = \begin{pmatrix} \langle \sigma_x \rangle_\psi \\ \langle \sigma_y \rangle_\psi \\ \langle \sigma_z \rangle_\psi \end{pmatrix} \\ \text{or: } \begin{pmatrix} \alpha \\ \beta \end{pmatrix} &\mapsto \begin{pmatrix} \alpha\bar{\beta} + \beta\bar{\alpha} \\ i(\alpha\bar{\beta} - \beta\bar{\alpha}) \\ |\alpha|^2 - |\beta|^2 \end{pmatrix} \end{aligned}$$

Generally, given a smooth manifold M , a principal $U(1)$ -bundle is a fiber bundle $p: E \rightarrow M$ with a group action $U(1) \times E \rightarrow E$ such that $\forall e \in E, g \in G : e \in p^{-1}(x) \Rightarrow g \cdot e \in p^{-1}(x)$ and the action on each fiber is free and transitive. In our case $M = S^2$.

0.4 Examples of adjunctions

$Unif$ is category of uniform spaces and uniformly cts functions, $CUnif_{T_2}$ is category of complete hausdorff uniform spaces, $\hat{A}$ is completion of A .

$$Hom_{Unif}(A, Y) \cong Hom_{CUnif_{T_2}}(\hat{A}, Y)$$

0.5 Homeomorphism doesn't preserve completeness

Completeness is not a topological invariant. The spaces $(0, 1)$, $\mathbb{R}$ with Euclidean topology are homeomorphic via

$$h: \mathbb{R} \rightarrow (0, 1), \quad h(x) = \frac{\tanh(x) + 1}{2}$$

$\mathbb{R}$ is complete, $(0, 1)$ is not.

0.6 Frechet space that is not normable

Consider $L^1_{loc}(\mathbb{R}^d)$ given the F topology generated by seminorms

$$|f|_n = \int_{\|x\| \leq n} |f(y)| dy$$

Suppose there is a norm $\|\bullet\|$ generating the same topology. Then $\exists C > 0, C_n > 0, n \in \mathbb{N}$ such that

$$\|\bullet\| \leq C|\bullet|_N, \tag{1}$$

$$|\bullet|_n \leq C_n \|\bullet\| \tag{2}$$

from this we get $|\bullet|_{N+1} \leq C_{N+1} C |\bullet|_N$, which is clearly wrong for the function $f = \chi_{\{N < \|x\| < N+1\}}$.

To prove (1), for contradiction, assume

$$\forall n, k \in \mathbb{N} \exists \phi_{n,k} \in L^1_{loc} : \|\phi_{n,k}\| > n |\phi_{n,k}|_k$$

define $\psi_n = \frac{\phi_{n,n}}{n |\phi_{n,n}|_n}$, then

$$\forall l \in \mathbb{N}, n \geq l : |\psi_n|_l \leq |\psi_n|_n = \frac{1}{n} \rightarrow 0 \quad n \rightarrow \infty$$

so $\psi_n \rightarrow 0$ in L^1_{loc} but $\|\psi_n\| > 1$, contradiction. Generally, for any Frechet space X where the seminorms are nondecreasing, the inequalities (1) and (2) hold, so if we can find elements $x \in X$ such that $|x|_{N+1}$ can be arbitrarily large and $|x|_N$ arbitrarily small, the space is not normable.

For $C_0^\infty[0, 1]$ where topology is generated by $|f|_n = \sup_{k \leq n} \|f^{(k)}\|_\infty$, the element $f(x) = \frac{1}{n^N} \sin(nx)$ (n large) has small $|\bullet|_{N-1}$ but large $|\bullet|_N$.

0.7 Banach space with no preduals

Suppose $X^* = L^1(0, 1)$. Then by Banach-Alaoglu, $B_X = \{f \in L^1(0, 1) : \|f\| \leq 1\}$ is w^* -compact convex, so by Krein-Milman $B_X = \overline{\text{aconv}} \text{ ext } B_X$. But there are no extreme points of B_X :

Suppose $f \in B_X$ is extreme. Since $\int_0^x |f(t)| dt$ is continuous in x ,

$$\exists s \in (0, 1) : \int_0^s |f(t)| dt = \frac{\|f\|_1}{2}$$

Now $g_1 := 2f\chi_{(0,s)}$, $g_2 := 2f\chi_{(s,1)}$ satisfy $f = \frac{g_1 + g_2}{2}$, $g_1, g_2 \in B_X$, $g_1, g_2 \neq f$.

0.8 Fourier series of continuous function does not converge

. Let for $f: [0, 2\pi] \rightarrow \mathbb{R}$

$$S_N f(x) = \sum_{-N}^N a_n e^{-inx} = \frac{1}{2\pi} \int_0^{2\pi} D_N(x-t) f(t) dt,$$

where D_N is the Dirichlet kernel:

$$D_N(x) = \frac{\sin\left(\frac{2N+1}{2}x\right)}{\sin\left(\frac{x}{2}\right)}.$$

It is well-known that

- for $f \in L^p(0, 2\pi)$, $S_N f \rightarrow f$ in $L^p(0, 2\pi)$, $1 < p < \infty$,
- for $f \in BV[0, 1]$, $S_N f \rightarrow f$ pointwise on $[0, 2\pi]$,
- for $f \in AC[0, 1]$, $S_N f \rightrightarrows f$ on $[0, 2\pi]$.

Fix $x \in [0, 2\pi]$, $N \in \mathbb{N}$. Then define a signed Radon measure

$$\Phi_{x,N}: C[0, 2\pi] \rightarrow \mathbb{R}, \quad \Phi_{x,N} f = S_N f(x)$$

it's norm is

$$\|\Phi_{x,N}\|_{(C[0,2\pi])^*} = \frac{1}{2\pi} \int_0^{2\pi} |D_N(s)| ds$$

But

$$\frac{1}{2\pi} \int_0^{2\pi} |D_N(s)| ds \geq \frac{1}{2\pi} \int_0^{2\pi} \frac{\sin\left(\frac{2N+1}{2}s\right)}{\frac{s}{2}} ds \rightarrow \infty$$

Therefore $\{\Phi_{x,N}\}_{N \in \mathbb{N}}$ is not bounded in $(C[0, 2\pi])^*$. Using uniform boundedness principle (and that x was arbitrary) we get

$$\forall x \in [0, 2\pi] \exists f \in C[0, 2\pi]: S_N f(x) \text{ diverges as } N \rightarrow \infty \quad (3)$$

It gets worse. Denote $\mathcal{P} \subset L^1(0, 2\pi)$ the set of trigonometric polynomials. Clearly, $\mathcal{P}$ is dense. Let f be a function "bad" at x from (3). Then $\mathcal{P} + f := \{p + f : p \in \mathcal{P}\}$ is also dense and all $p + f \in \mathcal{P} + f$ are "bad" at x in the sense of (3). Therefore

$$\forall x \in [0, 2\pi] \exists \mathcal{D} \subset C[0, 2\pi] \text{ dense such that } \forall f \in \mathcal{D}: S_N f(x) \text{ diverges as } N \rightarrow \infty$$